



University of Pisa
Data Science & Business Informatics

DM 2 - Data Mining Project

Child Mind Institute (CMI) Dataset – Problematic Internet Use

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AY 2025–2026

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1 Introduction

This report presents an analysis of an extended and modified version of the well-known **Child Mind Institute (CMI) - Problematic Internet Use dataset**. The study aims to investigate the relationship between physical activity and fitness related data in children and adolescents and the early signs of problematic Internet use. The objective is to identify relevant patterns that may support the early detection of at-risk situations and contribute to the prevention of unhealthy digital habits.

2 Data Understanding and Preparation of Tabular Dataset

2.1 Data Semantics and Dataset Structure

The tabular dataset contains **8460** records and **82** attributes, including demographic variables, seasonal information, physical measurements, fitness assessment scores, bioelectrical impedance indicators, physical activity questionnaire scores, sleep-related variables, internet-use questionnaire items, and the target variable *sii*, which classifies participants into different levels of problematic internet use severity. As a first step, we analyzed the semantic meaning of each attribute using the provided `data_dictionary`. This allowed us to distinguish the role of each field and to group variables according to their source instrument and type of information.

We then inspected the column data types and found that the dataset was mainly stored as `float64`, `int64`, and `object`, which are not always appropriate or memory-efficient given the variables’ ranges and semantics. We therefore optimized the `dtypes` by downcasting numerical columns to smaller nullable integer or floating-point types when possible, converting binary variables to Boolean, and low-cardinality text variables to categorical types. This reduced memory usage from 8.96 MB to **2.09 MB**, a **76.64%** reduction, without altering the information content.

After this preliminary structural inspection, we analyzed the empirical distributions of all variables. For continuous numerical attributes, we combined **histograms**, index-value **scatter plots**, and **boxplots** in order to inspect both the marginal distribution and the behavior of values along the row order, while preserving missing values as visible gaps in the plots. For categorical and Boolean variables, we used **frequency plots** and **categorical summaries**, again preserving the row-wise perspective. This combined visualization strategy was useful not only for identifying distributional shapes, extreme values, and missingness patterns, but also for detecting possible distributional changes along the dataset index.

2.2 Preliminary Data Quality Assessment

This analysis revealed a clear discontinuity around row **index 3960**. Several variables showed different empirical behavior before and after this point, especially in terms of value granularity, distributional shape, and consistency among related physical variables. Considering that the project guidelines specify that the provided dataset is a modified version of the original CMI dataset with additional synthetic samples and considering that the original CMI training dataset contains 3960 records, this discontinuity strongly suggests that the first 3960 rows correspond to the original data, while the subsequent rows correspond to the added **synthetic portion**. For this reason, subsequent data-quality checks were often performed separately on the pre-3960 and post-3960 subsets, in order to avoid mixing potentially authentic and synthetic distributions during semantic validation and cleaning.

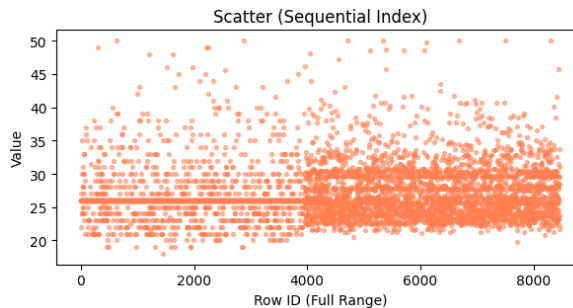


Figure 1: Scatter plot showing the distribution of values across the sequential row index.

The preliminary exploration also highlighted several **data-quality issues** that required a semantic rather than purely statistical inspection. In particular, the combination of the plots revealed the presence of outlying values. Some of them could be interpreted as extreme but still plausible observations, while others were clearly incompatible

with the expected physical or clinical meaning of the corresponding variable. For this reason, outliers were not treated uniformly: we distinguished between **statistical outliers**, which may still represent valid rare cases, and semantically **invalid values**, which violate the admissible range of the measurement (for example, zero or negative values were detected in variables for which such values are not physically possible). This was especially relevant for physical, clinical, and body-composition attributes, where the admissible value range is constrained by biological plausibility.

A second major issue was the large amount of **missing data**. Overall, **99.62%** of the records contained at least one missing value. Missing values represented **33.04%** of all cells, with an average of 27.09 missing attributes per row. The issue was not uniformly distributed across the dataset: in the first 3960 rows missing cells accounted for 24.93% of the data, while in the second portion the percentage of missing cells increased to 40.17%. At the column level, missingness affected different semantic groups with different intensity (For example the strongest increase was observed for the PCIAT-related variables, whose missingness rose from 31.82% in the first portion to 99.69% in the second one); these results confirmed that missing data management was a central part of the preparation pipeline rather than a marginal cleaning step.

2.3 Discretization, Semantic Inconsistencies, Outliers: Our choices

To reduce the difference between the original and synthetic portions of the dataset, we harmonized the **discretization** of the variables that showed different discretization patterns before and after row 3960. For each variable, specific rules were applied to map the values to the same measurement grid and make the two portions of the dataset consistent.

Semantic validation was then performed by defining admissible domains for each variable. Continuous attributes were checked against broad operational plausibility intervals derived from the data dictionary, the measurement semantics, and physiological constraints. Ordinal and categorical attributes were instead validated against their explicitly permitted code sets. Values that violated these constraints were masked as **NaN** and subsequently handled within the missing-value procedure; in total, 869 cells ($< 0.13\%$) were converted to **NaN** through this procedure. Observations that were unusual from a statistical perspective but still fell within their admissible semantic domain were left unchanged; these values may represent valid **rare cases** and were therefore reserved for the subsequent multivariate outlier-detection analysis rather than being removed during semantic cleaning.

After the first semantic cleaning step, we performed more detailed consistency checks on individual variables and on relationships between semantically connected attributes. Three main issues were identified:

In the first portion of the dataset, **Physical-BMI** was almost perfectly consistent with the standard formula for imperial units, $703 \times \text{weight (lb)} / \text{height (in)}^2$, however, after row 3960 this relationship was no longer reliably preserved. For this reason, the original BMI column was not used directly, and a new variable, **Physical-BMI_recalc**, was computed from **Physical-Height** and **Physical-Weight**.

PAQ_A and **PAQ_C**, which refer respectively to the PAQ test for adolescents and children, often contained values in the same row, contradicting their intended mutually exclusive age-based use. According to the questionnaire documentation found online, we used 14 years as the threshold: for participants aged 14 or older, only the adolescent version was retained, while for younger participants only the children version was retained; values in the age-inconsistent column were therefore set to missing. After this age-based validation, the two variables were merged into a single **PAQ_Total** variable.

Physical-Systolic_BP and **Physical-Diastolic_BP** were checked jointly, since systolic blood pressure must be greater than diastolic blood pressure by definition. The five rows violating this constraint were considered unreliable, and the corresponding blood-pressure values were set to missing.

2.4 Correlations

We computed a **Pearson correlation** matrix to identify possible linear redundancies among numerical variables and to guide the attribute-reduction step. Several strong correlations emerged, especially within homogeneous groups of measurements; this was particularly evident in the **BIA** block, where many body-composition indicators showed correlations above **0.8**, confirming that several of them carried largely overlapping or derivable information. We also observed a very high correlation between **PCIAT-PCIAT_Total** and *sii* of about **0.9**.

2.5 Missing Values Imputation

Given the large amount of missing data, missing-value imputation was treated as a **model-selection** problem rather than as a simple preprocessing step, since the basic median imputation strategy was not sufficient and was therefore used only as a baseline to outperform.

Since almost all columns contained missing values, we initially conducted an exploratory comparison using the Python implementation of **IterativeImputer**, which iteratively models each incomplete variable as a function of the others and repeatedly updates the missing estimates. We compared three strategies: **median** imputation as a baseline, the native **IterativeImputer** based on **BayesianRidge**, which is suitable for approximately linear relationships among correlated variables, and a customized **IterativeImputer** using Random Forest. **Random Forest** was included as a complementary alternative because, unlike a linear imputer, it can capture nonlinear patterns and interactions between predictors, allowing us to compare methods based on different modelling assumptions.

The comparison was performed through an artificial-masking validation scheme: for each method, **10%** of the originally observed values were hidden using the same masks and three random seeds, and the imputation error was computed only on these hidden cells. The evaluation used **MAE** and **RMSE**, and in particular we chose the latter, to penalize larger errors more strongly, normalizing it by the interquartile range of each variable, which allowed errors from attributes measured on different numerical scales to contribute comparably (ordinal variables were additionally evaluated through exact accuracy and within-one accuracy). This benchmark showed that the iterative Random Forest imputer outperformed both the median baseline and the Bayesian Ridge imputer, with a **macro-NRMSE** of **1.0901**, compared with 1.2386 for Bayesian Ridge and 1.3801 for median imputation, and also showed superior performance on the ordinal variables.

However, the standard **IterativeImputer** was mainly suitable as a preliminary benchmark, since it used the same regression estimator and hyperparameter configuration for all incomplete variables within a common iterative procedure. For this reason, we tested a **single-target** Random Forest imputation pipeline, which allowed each variable to be handled separately through target-specific hyperparameter tuning and the use of either regression or classification according to the variable type (for ordinal variables, since the categories followed a meaningful order rather than representing unrelated labels, we also compared classification with regression followed by clipping and discretization; since their performances were similar, **RandomForestClassifier** was retained because it better preserved the discrete nature of the targets).

The imputation matrix included 8460 rows and 38 selected numerical features (the columns that had already been selected for removal at this stage, and that did not provide any benefit to the imputation process, were not imputed), with **66856 missing cells**; **Age** and **Sex** were complete and used only as predictors, while the remaining 36 variables were imputed sequentially in ascending order of missingness, from the column with the fewest missing values to the one with the most. **PCIAT-PCIAT_Total** (and, of course, also *sii*) was not used as a predictor during the imputation procedure, and **PCIAT-PCIAT_Total** was not imputed itself. This choice was made to avoid any possible leakage in the subsequent classification of *sii*, the target variable, given their high correlation (0.9). For the same reason, **PCIAT-PCIAT_Total** remained a suitable **candidate target** for subsequent regression analyses, since imputation-induced bias on this variable was also avoided.

For each target column, the same artificial validation mask was first used to estimate column-specific error; then, the final model was trained on all rows where that target was observed and used to predict only its real missing values. Already imputed variables were available as predictors for later steps, while predictors still containing missing values were handled inside the pipeline with `SimpleImputer(strategy="median")`.

Configuration	Continuous variables	Ordinal variables
Estimator	<code>RandomForestRegressor</code>	<code>RandomForestClassifier</code>
Number of trees	100	100
Cross-validation	3-fold KFold	3-fold StratifiedKFold
Optimization criterion	RMSE	MAE between ordered categories
Hyperparameter tuning with:	GridSearchCV with <code>max_depth</code> $\in$ {None, 15}; <code>min_samples_leaf</code> $\in$ {1, 3}; <code>max_features</code> $\in$ {sqrt, 0.5, 1.0}	

Table 1: Target-specific configuration of the single-target Random Forest imputation pipeline.

After prediction, column-specific post-processing was applied to restore the correct measurement scale and admissible domain, ensuring that imputed values respected the expected integer, ordinal, and discretized structure of the original variables. The final single-target Random Forest method achieved the best overall performance, with a **macro-NRMSE** of **1.0655**, improving also over `IterativeImputer` with Random Forest (1.0901) while providing greater detail and control over the results of individual attributes, and so reducing the error (the overall normalized magnitude) by 22.8% compared with median imputation, our baseline. It also achieved the best performance on ordinal variables, with an exact accuracy of 62.21% and a within-one accuracy of 90.92%.

2.6 Column removal and attribute reduction

This reduction was guided not only by correlation analysis, although all columns identified as responsible for correlations above the 0.8 threshold were removed, but also by the semantic interpretation of the variables, in order to reduce multicollinearity without discarding independent information. For coherence and clarity, all removals are reported together in this section, although they were performed at different stages of the preparation process.

- **Feature engineering-derived removals:** As described in the previous paragraph, the original `Physical-BMI`, `PAQ_A-PAQ_A_Total`, and `PAQ_C-PAQ_C_Total` columns were removed and replaced. `Fitness_Endurance-Time_Mins` and `Fitness_Endurance-Time_Sec`, whose missing values did not always occur in the same rows, were imputed separately before being combined into a single variable, `Fitness_Endurance-Time_Total_Sec`, by multiplying the minutes by 60 and adding the seconds. Moreover, the left and right sit-and-reach measurements, `FGC-FGC_SRL` and `FGC-FGC_SRR`, were replaced by two summary variables: `FGC-FGC_SR_Mean`, capturing the average flexibility level, and `FGC-FGC_SR_Asymmetry`, capturing the absolute difference between the two sides.
- **Temporal variables:** Seasonal variables described only the season in which each group of measurements was collected, rather than a direct subject-specific or discriminative characteristic; moreover, they contained a large amount of missing data. Therefore, they were considered not sufficiently informative for the subsequent analyses and were discarded before imputation.
- **Redundant or derivable variables:** Several variables were removed because they were direct transformations or aggregations of information already available in other columns. All of these variables, except the BIA indicators, were removed before imputation because their missingness coincided with that of the retained source variables, so they could not provide additional information for the imputation process.
 - To reduce dimensionality while still preserving the aggregated information, all individual PCIAT item columns were removed, and only `PCIAT-PCIAT_Total` was retained as the global score.
 - To avoid redundancy, the `FitnessGram` zone variables were removed because they represent binary or ordinal fitness classifications, derived from the corresponding raw `FitnessGram` measurements.
 - `SDS-SDS_Total_T`, a standardized T-score version, was removed because `SDS-SDS_Total_Raw` was

retained as the direct questionnaire score result.

- Several BIA variables were discarded because they were highly redundant body-composition indicators, often derived from formulas, normalizations, or decompositions involving other BIA and anthropometric measurements. The removed BIA variables were: BIA-BIA_BMI, BIA-BIA_LST, BIA-BIA_TBW, BIA-BIA_LDM, BIA-BIA_BMR, BIA-BIA_DEE, BIA-BIA_FFMI, BIA-BIA_FMI, BIA-BIA_FFM, and BIA-BIA_SMM.

2.7 Final Tabular Dataset

Finally, the `id` column was also removed because it is only an identifier and does not represent a predictive attribute, resulting in **30 columns** in total (`Basic_Demos-Age`, `Basic_Demos-Sex`, `CGAS-CGAS_Score`, `Physical-Height`, `Physical-Weight`, `Physical-BMI_recalc`, `Physical-Waist_Circumference`, `Physical-Diastolic_BP`, `Physical-HeartRate`, `Physical-Systolic_BP`, `Fitness_Endurance-Max_Stage`, `Fitness_Endurance-Time_Total_Sec`, `FGC-FGC_CU`, `FGC-FGC_GSND`, `FGC-FGC_GSD`, `FGC-FGC_PU`, `FGC-FGC_TL`, `FGC-FGC_SR_Mean`, `FGC-FGC_SR_Asymmetry`, `BIA-BIA_Activity_Level_num`, `BIA-BIA_BMC`, `BIA-BIA_ECW`, `BIA-BIA_Fat`, `BIA-BIA_Frame_num`, `BIA-BIA_ICW`, `PAQ_Total`, `SDS-SDS_Total_Raw`, `PreInt_EduHx-computer_int_hoursday`, `PCIAT-PCIAT_Total`, `sii`). After removing it, we checked for complete duplicate rows in the final modelling dataset. We found 101 duplicate records, reducing the dataset from 8460 to **8359 rows**. These duplicates were dropped to avoid biases in the subsequent analyses, especially in tasks such as outlier detection, and to prevent the same observation from potentially appearing in both training and test sets during predictive modelling.

3 Advanced Data-Preprocessing

3.1 Outlier Detection

This section presents the anomaly detection analysis performed on the final CMI tabular dataset. The target/leakage columns `sii` and `PCIAT-PCIAT_Total` were excluded before outlier detection, because `PCIAT-PCIAT_Total` is the raw score from which `sii` is derived. The remaining **28 numerical features** were standardised with `StandardScaler()` and used as the unsupervised feature space.

3.1.1 Outlier Detection Methods

Five methods were used, each belonging to a different algorithmic family. Initially, ABOD was tested, but was then discarded as its scores were numerically affected by near-duplicate observations; using it would have required either removing non-erroneous records or adding artificial jitter to make the algorithm stable.

For all used methods, higher outlier scores indicate greater anomalousness and each detector contributes its own top 1% of records (`CONTAMINATION = 0.01`), corresponding to **84 observations**. Table 2 summarises the family, key **hyperparameters** (consistent with the lab defaults except for `n_estimators=200` that we validated through a stability check across different tree counts and `n_neighbors=20` to be consistent with LOF), and average anomaly scores assigned to outliers and inliers, separation is computed as the ratio between the mean outlier score and the mean inlier score.

Method	Family	Key Hyperparameters	Top-1% Outliers	Mean Score Outliers	Mean Score Inliers	Separation
LOF	Density-based	<code>n_neighbors=20</code>	84	8.040	1.340	6.00×
LODA	Projection / histogram ensemble	<code>n_random_cuts=100</code> , <code>n_bins=10</code>	84	0.031	0.021	1.51×
Isolation Forest	Tree ensemble	<code>n_estimators=200</code> , <code>max_samples=auto</code>	84	0.580	0.420	1.38×
KNN	Distance-based	<code>n_neighbors=20</code> , <code>method=largest</code>	84	9.739	3.688	2.64×
CBLOF	Clustering-based	<code>n_clusters=8</code>	84	11.735	4.101	2.86×

Table 2: Comparison of the five outlier detection methods used in the final notebook.

To make the **Isolation Forest** output more interpretable, we inspected the most anomalous observation according to this detector. The path followed by this observation ends after only **7 splits**, showing that the observation can be separated from the rest of the data with few random partitioning steps, this is coherent with the Isolation Forest mechanism, where more anomalous points tend to have shorter isolation paths.

3.1.2 Consensus Voting and Overlap Analysis

The five top-1% lists were combined through a consensus vote, all the flagged points have been compared and studied. A systematic analysis, done to determine an appropriate threshold, reveals that **no subject** is flagged by all **five** methods (the maximum number of votes observed is 4 out of 5) as shown in Table 3.

Votes	Exact records	Cumulative records	Cumulative %
4	20	20	0.24%
3	31	51	0.61%
2	55	106	1.27%
1	137	243	2.91%

Table 3: Consensus vote distribution across LOF, LODA, Isolation Forest, KNN and CBLOF

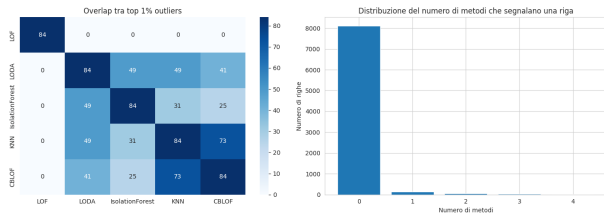


Figure 3: Pairwise overlap matrix and cumulative vote distribution.

views of the data.

The final threshold was set to at least **2 votes out of 5**. A threshold of 1 would correspond to the union of all methods and would remove 243 records (2.91%), including many single-method detections, while a threshold of 3 would remove only 51 records (0.61%) and would discard 55 two-method agreements. The ≥ 2 rule therefore keeps only records confirmed by at least two independent assumptions, while avoiding excessive removal.

Because class `sii = 3` is rare and clinically more delicate, the consensus outliers belonging to this class were inspected separately before final removal, showing that only **3 of the 86** severe cases were removed by the ≥ 2 rule, corresponding to 3.5% of the severe class. The flagged cases were incoherent in the empirical distribution within the record features, especially in body composition, anthropometric and functional variables. The most critical severe case of them was flagged by four different detection criteria (LODA, Isolation Forest, KNN and CBLOF) and shows several simultaneous deviations from the "standard" ranges. The other two, confirmed by LODA and Isolation Forest, showed a similar pattern of multiple anomalies. For these reasons, the records were not protected from removal solely because they belong to `sii = 3`: retaining them would introduce noise that could distort the classifier's decision boundaries more than it would help the model learn the severe class.

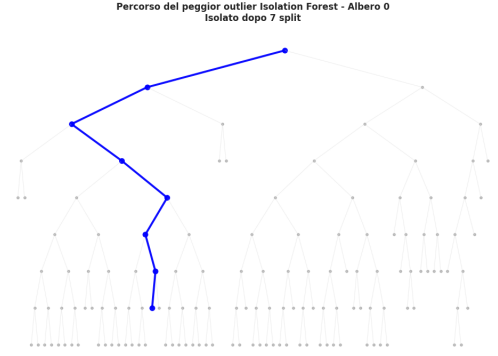


Figure 2: Single-tree Isolation Forest path for the most anomalous record.

From Figure 3 we can see that LOF has **zero overlap** with the other four methods, and we deduced that its alerts correspond to local density anomalies rather than globally extreme feature profiles. By contrast in the heatmap, the four non-LOF methods overlap substantially, our decision was then to remove the flags that represented cross-family agreement among projection-based, tree-based, distance-based and clustering-based

3.1.3 Dimensionality Reduction and Visualisation

Two dimensionality reduction techniques were applied to visualise the spatial distribution of outliers in the 28-dimensional feature space: **Principal Component Analysis (PCA)** and **t-distributed Stochastic Neighbour Embedding (t-SNE)**. PCA was used as the main linear projection: the first component explains 25.42% of the variance, the first two components **32.51%**, reaching 80% of explained variance requires **15 components**, while reaching 90% requires **20 components**. t-SNE was added as a non-linear local view, with **perplexity = 30** and **n_iter = 1000**.

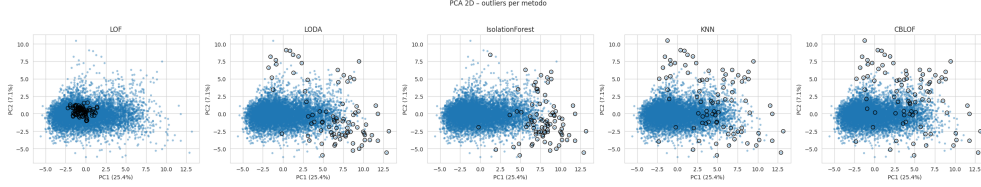


Figure 4: 2D PCA projection of the outliers identified by each method.

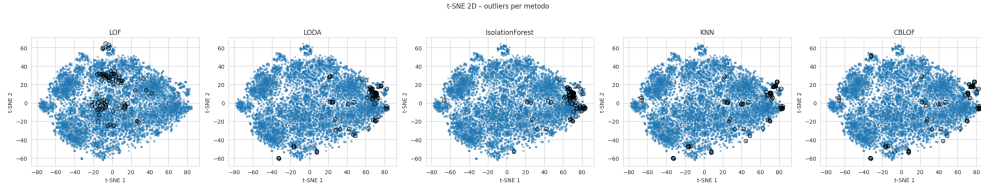


Figure 5: 2D t-SNE projection of the outliers identified by each method.

The PCA projection places many consensus outliers on the periphery of the cloud, consistently with global feature extremity. The t-SNE view shows that many of the same observations also occupy sparse local regions or small edge clusters.

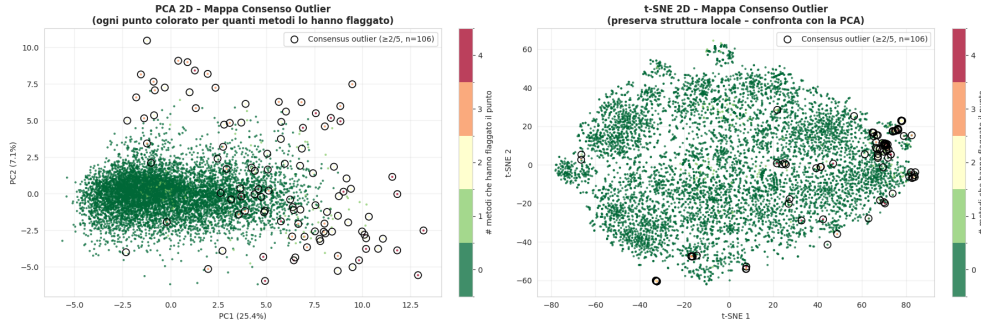


Figure 6: Consensus outlier map: each point is coloured by the number of methods that flagged it, consensus outliers ($\geq 2/5$, $n=106$) are circled in black

3.1.4 Handling Outliers

The final treatment removes **106 records**, reducing the dataset from **8,359** to **8,253 records**. The distribution of the target variable **sii** after removal is largely preserved in relative terms: None (**5,678**, from 5,739), Mild (**1,560**, from 1,586), Moderate (**932**, from 948), and Severe (**83**, from 86).

3.2 Imbalanced Learning

Imbalanced learning techniques were applied to a new categorical variable, **sii_Binary_Tabular**, engineered by binarizing the **sii** score into **'Mild'** (0-1, no problematic use) and **'Severe'** (2-3, severe internet addiction risk).

To prevent data leakage and artificial inflation, the original `sii` and its source variable `PCIAT-PCIAT_Total` were removed because either would provide direct target information and artificially inflate predictive performance. The feature space was normalized using `MinMaxScaler()`, which yielded cleaner visualizations for the PCA and better results than `StandardScaler()`. Figure 7 shows the initial class distribution.

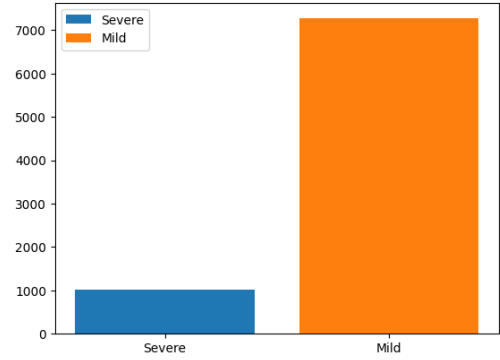


Figure 7: `Sii_Binary_Tabular` classes frequencies.

A **Decision Tree (DT) classifier** trained on the original dataset served as a baseline. It showed a strong bias toward the majority class (Mild), yielding poor recall and precision for the minority class (Severe). Several resampling techniques were explored to mitigate this issue.

3.2.1 Undersampling

Undersampling aims to balance the class distribution by reducing the size of the majority class. Four different techniques were implemented and evaluated: **Random Undersampling**, **Tomek Links**, **Edited Nearest Neighbours (ENN)**, **Cluster Centroids**. Their post-transformation distributions (via PCA) are shown in Figure 8.

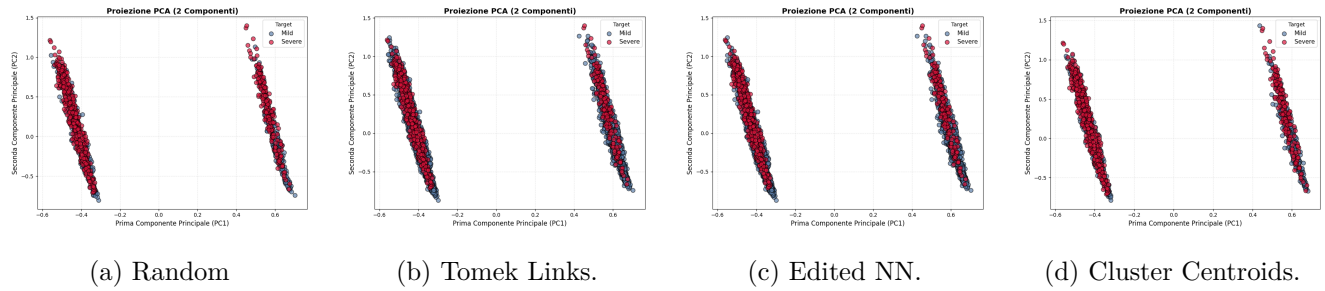


Figure 8: PCA visualization on two principal components of the `Sii_Binary` classes, following the application of different undersampling techniques.

For a controlled comparison, the same Decision Tree configuration and the same untouched test set were used across all resampling methods. Random Under Sampling and Cluster Centroids (threshold set to the minority class size) severely degraded overall accuracy; though RUS improved minority recall, Cluster Centroids destroyed performance. ENN slightly cleared the decision boundary by removing noisy majority samples, marginally improving minority recall and F1-score while preserving high accuracy. Tomek Links only removed borderline instances, yielding results nearly identical to the baseline. Thus, **standalone undersampling proved inadequate**.

3.2.2 Oversampling

Various techniques were also used for oversampling, such as **Random Oversampling**, **SMOTE**, and **ADASYN**. In this case, the results were very similar for all three methods, with minimal differences in all cases between the initial metrics and the post-transformation metrics, where we see a slight improvement for the minority class.

3.2.3 Balancing at algorithm level

An attempt was then made to balance the class weights directly at the **algorithm level**. In this case as well, a **Decision Tree** classifier was used. Through a grid search, various class weights and different values for `min_samples_leaf` were tested. The best parameter configuration found was: `'Mild':1, 'Severe':2, 'min_samples_leaf':3`. However, the results were disappointing, as the classifier did not significantly improve the

prediction of the minority class.

3.2.4 Combined Approach

A hybrid pipeline was designed to combine the benefits of **oversampling** and **boundary cleaning**. The data were first split into stratified training and test sets. `MinMaxScaler()` was fitted exclusively on the training data and then applied to both partitions. All under- and oversampling techniques were applied only to the training set, while the same untouched test set was used for all performance comparisons. **ADASYN** was used to expand the minority class from 710 to **5259 samples**, focusing on hard-to-learn overlapping regions. **ENN** was then applied as a filter, removing noisy samples to better define the decision boundary, reducing the Severe class to **5099 instances** while keeping the Mild class intact. ADASYN was chosen because, unlike uniform synthetic oversampling, it adaptively generates more samples around minority observations whose local neighbourhoods contain a larger proportion of majority-class examples, thereby focusing the oversampling effort on harder-to-learn regions.

	Precision		Recall		F1-score	
	before	after	before	after	before	after
Mild	0.89	0.91	0.89	0.69	0.89	0.78
Severe	0.22	0.19	0.22	0.52	0.22	0.28
Accuracy					0.80	0.67
Macro Avg	0.55	0.55	0.55	0.60	0.55	0.53
Weighted Avg	0.81	0.82	0.80	0.67	0.81	0.72

Table 4: Comparison of Decision Tree classification metrics across baseline and hybrid balancing strategies.

This pipeline improved minority recall at the expense of majority recall and overall accuracy (Table 4). The **ROC curves** (Figure 9) show a marginal improvement. The visual overlap in the projected feature space, together with the limited predictive performance, is consistent with substantial class overlap. This lack of separation explains the **high rate of false positives** (low precision) despite identifying more positive cases (higher recall).

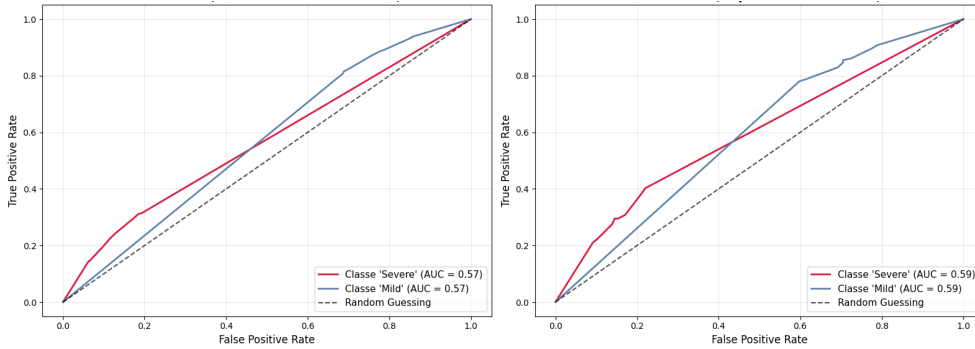


Figure 9: Visualization of the `Sii_Binary_Tabular` decision tree ROC curves, before (left) and after (right) combined approach of rebalancing techniques.

4 Advanced ML & XAI

In this module, a multi-class classification task was addressed using the **Severity Impairment Index (sii)** target variable, which measures the level of problematic internet use in children and adolescents. The dataset (28 features) was split into a training set of **5777 samples** (70%) and a test set of **2476 samples** (30%). To handle extreme class rarity, the original class 3 (83 records) was **merged** into class 2, resulting in three final classes heavily skewed toward the majority (Class 0: 3975, Class 1: 1092, Class 2: 710). All numerical variables were standardized via `StandardScaler`, excluding `PCIAT-PCIAT_Total` from the features used, to prevent bias. For each model, the detailed **Classification Report** was analyzed for each class of `sii`, including **Precision**, **Recall**, **F1-Score** metrics, **average values** (Macro and Weighted) and **ROC curve**; given the class imbalance,

hyperparameter selection was based on the macro-averaged F1-score (**macro-F1**) to assign equal importance to each class and therefore prevent the majority class from dominating the model-selection criterion.

4.1 Logistic Regression

The classifier was initially built with the parameters `class_weight='balanced'`. To improve performance, hyperparameters were optimized via `GridSearch` with Cross-Validation, testing $C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$ and solvers `'lbfgs'`, `'saga'`. The optimal configuration selected was `C:0.001`, `solver:lbfgs`.

As shown in Table 5, the model performs well on Class 0 with a higher precision (0.79) and a solid f1-score (0.71) but struggles significantly with Class 1. Class 2 exhibits a higher Recall (0.53) at the expense of Precision (0.25).

Table 5: Classification Report - Logistic Regression

Classe (sii)	Prec.	Rec.	F1	Sup.
0	0.79	0.64	0.71	1703
1	0.25	0.24	0.25	468
2	0.25	0.53	0.34	305
Accuracy			0.55	2476
Macro Avg	0.43	0.47	0.43	2476
Weighted Avg	0.62	0.55	0.58	2476

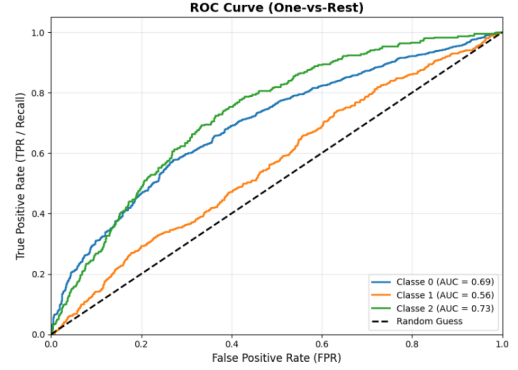


Figure 10: ROC Curves - Logistic Regression

The ROC curves (Figure 10) confirm a baseline discrimination capability above random guessing (**AUC = 0.69** for the class 0). The lower overall accuracy (55%) reflects a **trade-off**: the model sacrifices part of the majority-class recall while achieving substantially higher sensitivity to the class 2. The limited performance may reflect class overlap, imbalance and relationships not fully captured by a linear decision function; the following nonlinear models were therefore evaluated.

4.2 Support Vector Machines

For the SVM classifier, four kernel types were explored: **Linear**, **Radial Basis Function (RBF)**, **Polynomial**, and **Sigmoid**. All models were initialized with `class_weight='balanced'` to counteract the native dataset imbalance. For each kernel module, a targeted hyperparameter tuning phase was performed via `GridSearch` and `Cross-Validation`.

- **Linear Kernel:** The linear kernel defines a linear separating hyperplane in the original feature space. We tested the regularization parameter $C \in [0.001, 0.01, 0.1, 1, 10, 100]$.
- **Non-Linear RBF Kernel:** The RBF kernel implicitly maps the data into an infinite-dimensional space, which is ideal for capturing spherical patterns or complex non-linear groupings. Tested parameters: $C \in [0.1, 1, 10]$ and the kernel coefficient parameter γ (`gamma`) $\in ['scale', 'auto']$.
- **Non-Linear Polynomial Kernel:** This kernel evaluates interactions between features up to a specified combinatorial degree. Tested parameters: $C \in [0.1, 1, 10]$, the kernel coefficient parameter γ (`gamma`) $\in ['scale', 'auto']$, the polynomial degree `degree` $\in [2, 3]$, and the independent coefficient `coef0` $\in [0.0, 1.0]$.
- **Non-Linear Sigmoid Kernel:** Derived from the concept of artificial neural networks (hyperbolic tangent activation function), this kernel maps the data similarly to a two-layer perceptron. Tested parameters: $C \in [0.1, 1, 10]$, `gamma` $\in ['scale', 'auto']$, and `coef0` $\in [0.0, 1.0]$.

The best hyperparameters for each kernel were selected through cross-validation on the training set. The resulting **tuned SVM variants** were then evaluated on the held-out test set for final comparison. Table 6 summarizes the obtained aggregate metrics and the best hyperparameters found for each model.

Table 6: Detailed comparison of the performance and optimal hyperparameters of the SVM kernels.

Classe (sii)	Kernel Lineare			Kernel RBF			Kernel Polinomiale			Kernel Sigmoido		
	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1
0	0.79	0.64	0.71	0.81	0.66	0.73	0.80	0.67	0.73	0.81	0.55	0.65
1	0.24	0.26	0.25	0.27	0.35	0.31	0.28	0.33	0.30	0.21	0.24	0.23
2	0.25	0.48	0.33	0.23	0.37	0.29	0.24	0.39	0.29	0.19	0.49	0.27
Accuracy	0.55			0.57			0.57			0.48		
Macro Avg	0.43	0.46	0.43	0.44	0.46	0.44	0.44	0.46	0.44	0.40	0.43	0.38
Weighted Avg	0.62	0.55	0.58	0.64	0.57	0.59	0.63	0.57	0.60	0.62	0.48	0.52
Migliori Iperparametri (Grid Search)												
C	0.01			1			0.1			0.1		
gamma (γ)	-			auto			auto			scale		
degree	-			-			3			-		
coef0	-			-			1.0			0.0		

The balanced class weights increased the penalty associated with errors on minority-class observations, shifting the learned decision boundaries toward greater minority sensitivity. SVMs generally improve the recall of Class 1, but none of the tested kernels reaches the Class 2 recall obtained by Logistic Regression. However, this shift in the decision boundaries introduces a **precision–recall trade-off**, causing more Class 0 observations to be assigned to minority classes and, for some kernels, reducing the recall of the majority class.

The small advantage of RBF and Polynomial kernels suggests that some nonlinear structure may be present, but the limited differences from the linear kernel indicate that nonlinearity alone does not resolve the class-separation problem. The Linear kernel failed to draw clean boundaries around the intermediate psychological state (Class 1), which remained a persistent classification bottleneck across all models. Conversely, the Sigmoid kernel performed worst among the tested configurations, indicating that this kernel formulation was poorly suited to the present task under the explored hyperparameter settings.

A common pattern across all four models (although mitigated by the non-linear kernels) is the persistent difficulty in accurately classifying **Class 1**.

Ultimately, the **RBF Kernel** provided a marginally preferable compromise between generalization capability and minority sensitivity (Class 2 Recall = 0.37, Macro F1 = 0.44). This is further illustrated by the ROC and Precision-Recall curves in Figure 11, where Class 0 exhibits stable, dominant performance metrics.

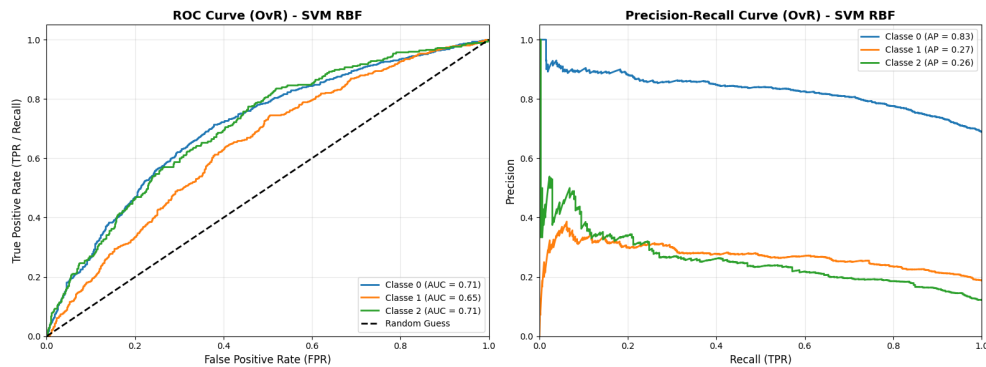


Figure 11: Visualization of the SVM ROC curves (left) and Precision_Recall curves (right).

4.3 Neural Networks

Initially, the `MLPClassifier` function from scikit-learn was used to build and train the neural network model. To prevent overfitting, the early stopping technique was employed, with the stopping criterion activated if the accuracy on the validation set did not improve by more than 10^{-4} for 20 consecutive epochs. This time, the hyperparameter tuning process was performed using `RandomizedSearchCV`, with the following search space defined for the model parameters: `'hidden_layer_sizes':[(256, 128, 64), (128, 64, 32), (64, 32), (32, 16), (16, 8), (16,), (8,)]`, `'alpha':[0.1, 0.01, 0.001]`, `'learning_rate_init':[0.01, 0.001, 0.0001]`, `'activation': ['tanh', 'relu']`. The best combination found was: `'learning_rate_init': 0.001`, `'hidden_layer_sizes': (32, 16)`, `'alpha': 0.01`, `'activation': 'tanh'`.

As shown in Table 7a, this unweighted model achieved the highest overall **accuracy (0.68)** due to its near-perfect performance on Class 0. However, because `MLPClassifier` lacks native class balancing, the network collapsed on the minority classes, yielding critically **low recall**. Target control was introduced by penalizing errors on minority

(a) MLP (scikit-learn)					(b) DNN (PyTorch)				
Classe (sii)	Prec.	Rec.	F1	Supp.	Classe (sii)	Prec.	Rec.	F1	Supp.
0	0.70	0.97	0.82	1703	0	0.81	0.62	0.70	1703
1	0.28	0.03	0.05	468	1	0.26	0.30	0.28	468
2	0.35	0.09	0.14	305	2	0.23	0.49	0.32	305
Accuracy			0.68	2476	Accuracy			0.54	2476
Macro Avg	0.44	0.36	0.33	2476	Macro Avg	0.44	0.47	0.43	2476
Weighted Avg	0.58	0.68	0.59	2476	Weighted Avg	0.64	0.54	0.57	2476

Table 7: Comparison of the classification reports for the baseline MLP model and the optimized Deep Neural Network implemented in PyTorch.

classes via a **weighted loss function** (`CrossEntropyLoss` with balanced weights). To prevent the network from memorizing the training data and to reduce overfitting risk, three combined techniques were integrated: **Dropout** (at 30% on each hidden layer), **Weight Decay** (L2 regularization in the Adam optimizer), and **Early Stopping** with automatic saving of the optimal checkpoint.

As illustrated in Figure 12, the Validation Loss reached its global minimum at **epoch 35** (1.0190), after which training was terminated at epoch 65 by the early stopping mechanism to prevent overfitting, restoring the optimal weights from epoch 35. The cost-sensitive and regularized PyTorch network substantially improved minority-class

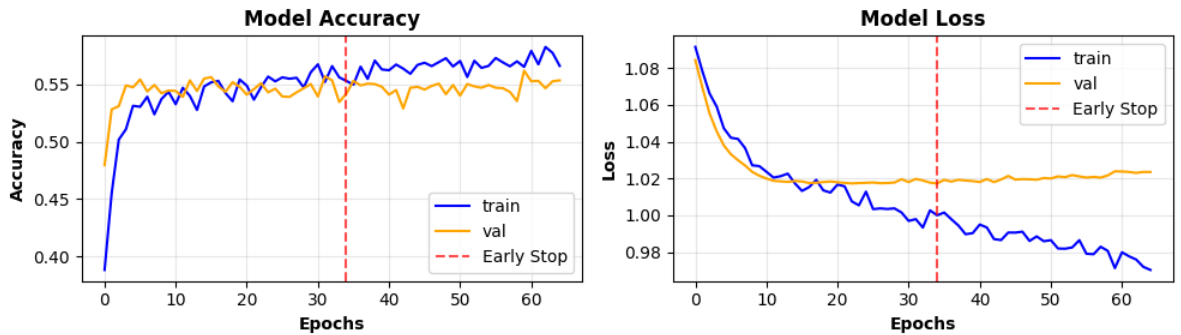


Figure 12: Visualization of the curves of Accuracy and the loss during the training

recognition compared with the unweighted `scikit-learn` MLP: Class 2 recall increased substantially to **0.49** and Class 1 recall reached **0.30** (Table 7b). Consequently, the overall accuracy dropped from 0.68 to 0.54; the reduction in accuracy reflects a deliberate **trade-off** rather than an unambiguous degradation: the weighted model sacrifices majority-class accuracy to achieve substantially higher sensitivity and a better macro-F1 across classes.

A subsequent test was performed using **SMOTE** to oversample the minority classes, and the neural network was

trained in the same manner as the previous one, but no improvements were observed.

Finally, a version of the Neural Network incorporating **Batch Normalization** was implemented: `nn.BatchNorm1d` layers were added between the linear layers and the **ReLU** activation functions. This technique normalizes the inputs of each hidden layer, accelerating convergence and making the network less sensitive to weight initialization. All other hyperparameters of the neural network remained identical to the previous models. Training stopped at epoch 76 via early stopping, and the best model was saved at epoch 46. The results, however, were in line with the previous model, yielding no evident improvements.

4.4 Ensemble Method

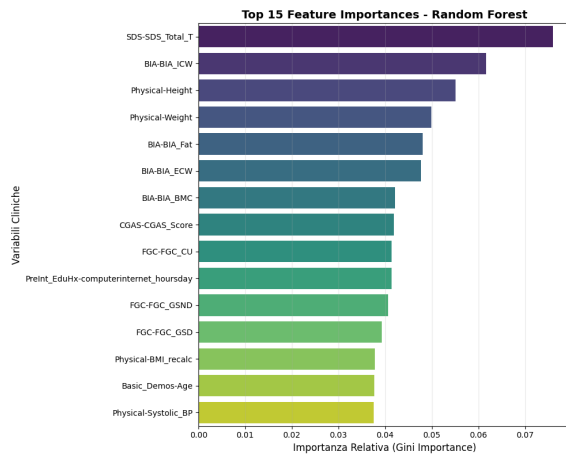
For the ensemble method, a **Random Forest** (RF) classifier was initially implemented, with hyperparameters tuned via **GridSearch** and **Cross-Validation**. The optimal configuration identified was: `'max_depth': 10`, `'min_samples_leaf': 2`, `'min_samples_split': 20`, `'n_estimators': 200`. Performance metrics are summarized in Table 8.

Subsequently, an **Extra Trees Classifier** was evaluated (`'n_estimators': 200`, `'min_samples_split': 5`, `'min_samples_leaf': 1`, `'max_depth': 20`). By utilizing random splits and training each tree on the entire dataset without bootstrapping, this model achieved a computational speedup compared to Random Forest while delivering nearly identical overall predictive performance.

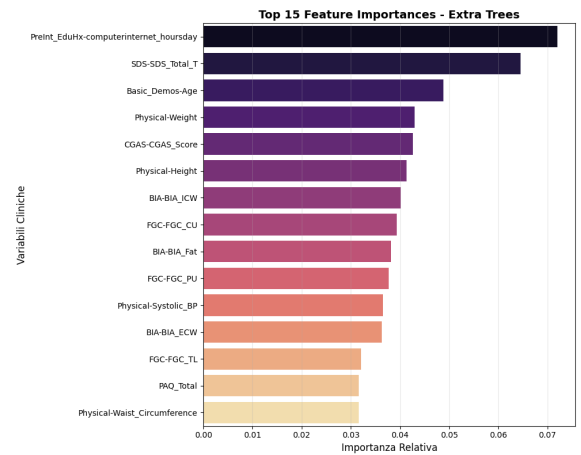
Table 8: Classification Report - Random Forest

Classe (sii)	Prec.	Rec.	F1	Sup.
0	0.78	0.80	0.79	1703
1	0.30	0.26	0.28	468
2	0.28	0.30	0.29	305
Accuracy			0.63	2476
Macro Avg	0.45	0.45	0.45	2476
Weighted Avg	0.63	0.63	0.63	2476

The Feature Importance of both models was evaluated using Mean Decrease in Impurity (MDI), as shown in Figure 13. For Random Forest (Figure 13a), the tree-splitting decisions are heavily driven by physical attributes, such as weight, height, and BIA measurements. A similar pattern emerges for the Extra Trees classifier (Figure 13b), where physical metrics still prominently contribute to the splits, though `PreInt_EduHx` emerges as a top feature, and no single variable dominates the feature space.



(a) Bar Plot of the most important features for Random Forest classifier.



(b) Bar Plot of the most important features for Extra Tree classifier.

Figure 13: Visual analysis of the feature importance of tree-based ensemble models.

4.5 Gradient Boosting Machines

The **XGBoost** algorithm was selected for the Gradient Boosting module due to its efficiency and flexibility. Hyperparameter tuning was performed via **GridSearch** with k -fold Cross-Validation under the same experimental pipeline defined previously: **PCIAT_Total** variable was removed, the dataset was split into training (70%) and test (30%) sets with stratification based on the target variable, and all features were standardized.

The optimal configuration identified balances control and regularization: a **learning rate**=0.2 coupled with **n_estimators**=300 and a **max_depth**=7. Overfitting was mitigated through stochastic regularization, setting **subsample**=0.6 and **colsample_bytree**=0.8, forcing the trees to train on only 60% of the instances and 80% of the available variables. Finally, conservative tree growth was enforced by setting **gamma**=0.2, alongside L2 regularization (**reg_lambda**=1) and **reg_alpha**=0.

Table 9: Classification Report - XGBoost

Classe (sii)	Precision	Recall	F1-Score	Support
0	0.73	0.93	0.82	1703
1	0.28	0.12	0.16	468
2	0.37	0.16	0.22	305
Accuracy			0.68	2476
Macro Avg	0.46	0.40	0.40	2476
Weighted Avg	0.60	0.68	0.62	2476

XGBoost achieved high overall accuracy (**0.68**) and somewhat higher minority-class precision, but remained strongly biased toward the majority class, as shown by the low minority recalls (Table 9). This behavior is further illustrated by the ROC and Precision-Recall curves in Figure 14.

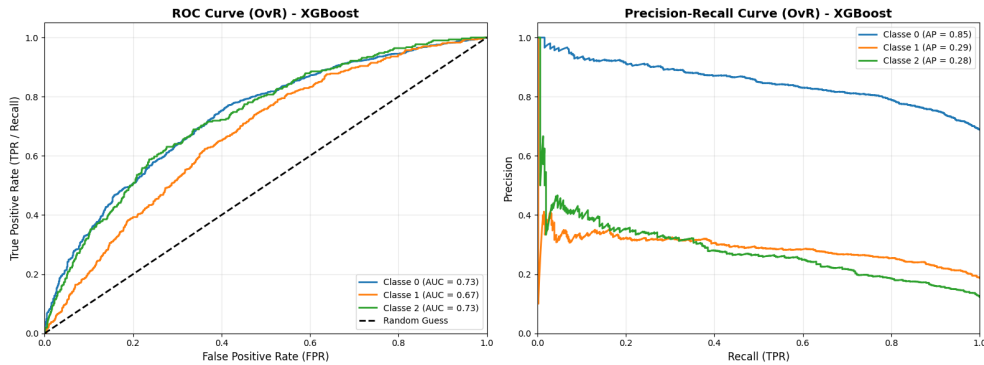


Figure 14: Visualization of the ROC curve and the Precision-Recall curve for the XGBoost classifier

A subsequent Feature Importance analysis revealed a broadly similar pattern to the Random Forest model, with physical and BIA attributes remaining the most indicative predictors without a single variable dominating the tree splits.

4.6 Advanced Classification Results

A comparison across all evaluated models trained to classify our target variable highlights a classic machine learning trade-off: balancing global accuracy against discrimination capability on minority, clinically relevant classes (**sii**=1 and **sii**=2), which are inherently penalized by the native dataset imbalance.

Two opposing predictive behaviors emerge from the metrics. The standard Multi-Layer Perceptron (**MLP**) and **XGBoost** achieved the highest raw accuracy (68%). However, this performance is biased by a strong tendency to predict the majority class (**sii**=0), resulting in a sharp drop in the **macro F1-Score** (0.33 for MLP and 0.40 for XGBoost) and critically low recall for the minority classes. Conversely, cost-sensitive and balanced models

(**Logistic Regression**, **SVM RBF**, and the PyTorch **DNN**) accepted a contraction in overall accuracy (ranging between 54% and 57%) to correct for data bias. This sacrifice led to a substantial sensitivity increase: the **DNN** recovered minority recall, raising the F1-scores of Class 2 and Class 1 to 0.32 and 0.28, respectively. Similarly, **Random Forest** achieved the best overall balance on the held-out test set, with the highest **macro F1-Score** of 0.45 and AUC values of 0.73 for Class 0, 0.65 for Class 1, and 0.71 for Class 2.

In conclusion, no single classifier dominates across all criteria. While XGBoost and the unweighted MLP achieve the highest reported raw accuracy, in a socio-health or psychological screening context aimed at the early detection of problematic internet use, **Random Forest** provides the best overall class-balanced performance according to **macro-F1**, with RBF and Polynomial SVMs following closely. Logistic Regression achieves the highest Recall for the merged problematic class, while the DNN is also notable for its relatively high recall on this class and for improved sensitivity to Class 1 compared with Logistic Regression. These gains in minority-class sensitivity come at the cost of lower overall accuracy and, in some cases, lower precision. In this domain, the primary objective shifts from minimizing global error to minimizing false negatives on borderline and severe classes, prioritizing sensitivity to potentially higher-severity cases.

5 Advanced Regression

This section addresses the multiple regression task on the target variable **PCIAT-PCIAT_Total**. The regression is deliberately performed only on records where the target variable is observed: Missing target values were not imputed, because doing so would make model training and evaluation depend partly on the target-imputation procedure rather than on observed outcomes; since we didn't use it as imputer either, it was the perfect candidate.

From the dataset that contains **8,253 records**, only **2,652** have an observed **PCIAT-PCIAT_Total** score and are used for regression, corresponding to **32.1%**; the remaining **5,601 records** are excluded from this task. An 80/20 split (resulting in **2,121 training records** and **531 test records**) was preferred over the 70/30 split used in classification, since the observed-target subset is substantially smaller. The leakage column **sii** is removed from the predictors because it is a discretized version of the same construct measured by **PCIAT-PCIAT_Total**.

5.1 Preprocessing and Model Design

No feature standardisation is applied, and this choice is intentional: all evaluated regressors are tree-based models, whose splits are based on ordered threshold comparisons on individual features. A monotonic rescaling such as **StandardScaler** would not add useful information and would mainly make the original feature scales less interpretable. The non-linear regression approaches that are compared are: **Random Forest**, **Gradient Boosting**, **XGBoost**, and **LightGBM**. For each method, a non-tuned baseline configuration is first evaluated (with library defaults), then **RandomizedSearchCV** with 30 sampled configurations, 5-fold cross-validation,

Model	Best tuned configuration
Random Forest	n_estimators=300; max_depth=10; min_samples_split=5; min_samples_leaf=10; max_features=0.5
Gradient Boosting	n_estimators=200; learning_rate=0.05; max_depth=3; min_samples_leaf=5; subsample=0.7
XGBoost	n_estimators=100; learning_rate=0.05; max_depth=3; min_child_weight=3; subsample=0.8; reg_alpha=0; reg_lambda=5
LightGBM	n_estimators=500; learning_rate=0.01; max_depth=-1; num_leaves=15; min_child_samples=10; subsample=0.7; reg_alpha=0; reg_lambda=0

Table 10: Best hyperparameter configurations selected by **RandomizedSearchCV**.

negative MAE as the optimisation score, is used to select a tuned configuration. Hyperparameter tuning optimises negative MAE, because MAE is directly interpretable in PCIAT score points and is less dominated by a few large residuals than RMSE. Table 10 shows the best hyperparameter combination found for each model. The final evaluation uses **MAE**, **RMSE**, and **R²** on the held-out test set.

A brief correlation screening is used only as a preliminary univariate check. It shows the strongest correlations with **Basic_Demos-Age** ($r = 0.410$), **Physical-Height** ($r = 0.406$), **PreInt_EduHx-computerinternet_hoursday** ($r = 0.362$), and **Physical-Weight** ($r = 0.362$): these variables are tied to physical growth and internet exposure, consistently with PCIAT scores tending to rise through adolescence. Notably, **SDS-SDS_Total_Raw**, which turns out to be one of the most important features for every tree-based model, has a linear correlation of only **0.229**, ranking ninth. This gap is informative: its predictive power could emerge mainly through non-linear interactions and threshold effects with other variables, which a simple Pearson coefficient cannot capture.

5.2 Baseline versus Tuned Models

Model	MAE	RMSE	R^2
Random Forest	14.233 → 14.142	17.528 → 17.452	0.251 → 0.258
Gradient Boosting	14.152 → 13.962	17.448 → 17.319	0.258 → 0.269
XGBoost	14.725 → 14.042	18.272 → 17.339	0.187 → 0.268
LightGBM	14.249 → 13.925	17.696 → 17.223	0.237 → 0.277

Table 11: Baseline and tuned test performance on observed PCIAT records ($n = 531$ test records). Arrows indicate baseline → tuned performance.

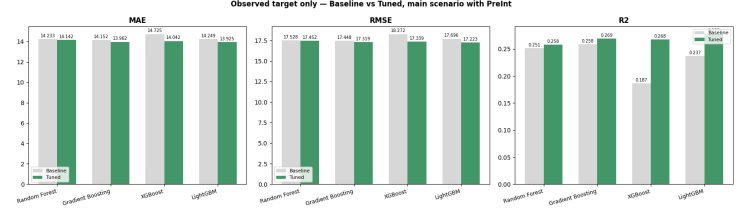


Figure 15: Baseline versus tuned comparison for MAE, RMSE and R^2 .

The differences between the tuned models are not dramatic, but they are consistent across the three evaluation metrics. The best final performance is achieved by **LightGBM**, with $MAE = 13.925$, $RMSE = 17.223$, and $R^2 = 0.277$, followed very closely by **Gradient Boosting**. This suggests that, in this specific regression task, boosting-based models are slightly more effective than Random Forest in capturing the non-linear structure of the data. A possible explanation is that the sequential correction of residual errors allows boosting methods to exploit weak but informative patterns distributed across demographic, physical, fitness-related, questionnaire-derived and behavioural variables. **Random Forest**, instead, appears more stable but less responsive to hyperparameter tuning, improving only marginally from $R^2 = 0.251$ to $R^2 = 0.258$.

XGBoost shows the largest relative gain after tuning, increasing from $R^2 = 0.187$ to $R^2 = 0.268$, which indicates that its baseline configuration was not well suited to this dataset but that the model can recover competitive performance when properly tuned and regularized. Nevertheless, it does not surpass LightGBM or Gradient Boosting. Overall, the best models explain approximately **27%** of the variance in the observed **PCIAT-PCIAT_Total** scores. This is a moderate result rather than a strong predictive performance, but it is methodologically meaningful because the regression is trained and evaluated only on genuinely observed target values, without relying on target imputation. In this sense, the results suggest that the available predictors contain a non-negligible signal related to problematic internet use, while also confirming that a large part of the variability of the **PCIAT** score remains unexplained by the available tabular features.

5.3 Feature Importance and Interpretation

Model	Top 5 features by impurity-based importance
Random Forest	Age; SDS Raw; PreInt hours; Height; Weight
Gradient Boosting	Age; SDS Raw; PreInt hours; Height; Weight
XGBoost	Age; Height; PreInt hours; SDS Raw; Weight
LightGBM	SDS Raw; Heart Rate; PreInt hours; CGAS Score; FGC Curl-Up

Table 12: Top-5 features according to the tuned models (variable names are abbreviated)

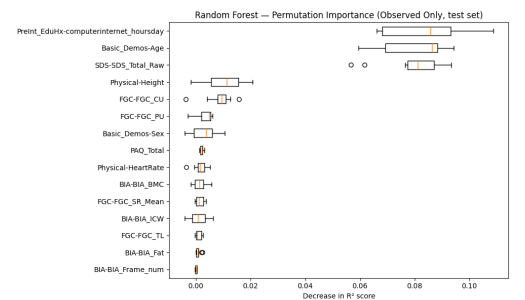


Figure 16: Random Forest permutation importance on the test set.

Age, SDS-SDS_Total_Raw, daily computer/internet hours, height, and weight appear repeatedly among the most important predictors across the four models by impurity-based importance (Table 12), consistent with the preliminary correlation analysis, though this should not be read as causal evidence. Permutation importance for Random Forest largely confirms this same set of predictors but reorders them: daily computer/internet hours emerges as the single most impactful feature, ahead of age, which ranked first by impurity-based importance. Weight, by contrast, does not appear among the top permutation-based features, suggesting that its high impurity-based importance may partly reflect correlations with other predictors or the greater number of available split points for a continuous variable, rather than an independent effect on the target.

5.4 Predicted vs Actual and Residual Analysis



Figure 17: Prediction diagnostics for the tuned regression models: predicted versus observed `PCIAT-PCIAT_Total` values on the left and residuals versus predicted values on the right.

The predicted-vs-actual scatter plots show a positive but visibly compressed relationship, coherently with tree-based ensembles trained on a noisy, moderately-sized target: predictions cluster closer to the mean of the target than the actual values do, systematically **under-estimating** the most severe cases (high `PCIAT` scores) and slightly **over-estimating** the lowest ones. This is an expected behaviour when the signal available in physical/fitness features is limited relative to the behavioural variability that `PCIAT` is actually trying to measure. This is also visible in the residual plots and is consistent with the moderate R^2 values. The available predictors capture part of the signal, but they do not fully explain the behavioural variability encoded in `PCIAT-PCIAT_Total`.

5.5 Conclusion

Using only records with an observed target avoids introducing artificial target values, although it reduces the training set and makes performance estimates less stable. All tuned models **improve over their baselines**, but the explained variance remains **limited**. This pattern is a realistic outcome for a regression problem where a complex psychometric score is predicted from demographic, physical, fitness, and questionnaire-derived measurements.

6 Explainability

The explainability analysis was carried out on the final **XGBoost black-box** classifier used for the three-class prediction of the Severity Impairment Index (`sii`), obtaining the target labels None (0), Mild (1), and Moderate+Severe (2). The objective of the XAI is to explain the behavior of the final model, including its limitations, trying to provide a global explanation to understand the overall model behavior and on examining individual cases to offer local, detailed interpretations of classification decisions.

The same preprocessing used for classification was preserved and the model achieved an accuracy of **0.68** and a macro-F1 of **0.40** on the test set: the model performs well on None, but both minority classes remain difficult, especially **Moderate+Severe**, whose recall is only 0.16. Because only a small number of Moderate+Severe test observations are correctly identified, explainability results must be read as explanations of the trained model, not as validated clinical rules. This is particularly important for **SHAP** and **LORE**: they can reveal which variables drive the model decision, but they do not prove causal mechanisms and they do not correct the limited recall of the high-risk class.

6.1 SHAP

SHAP was selected as the main explainability method because XGBoost is a tree-based model and can be explained efficiently through **TreeExplainer**. For the multiclass setting, SHAP returns one matrix of contributions for each class, with shape **2,476 × 28** in this run. SHAP values represent the **contributions** to the internal model output for a class, not direct changes in predicted probability; therefore, they should be interpreted in terms of direction and relative strength of the model evidence.

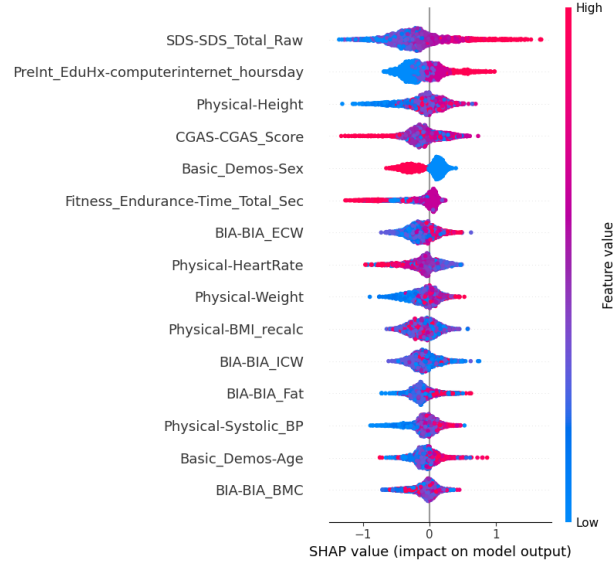


Figure 18: SHAP beeswarm plot for the Moderate+Severe class

Table 13: Feature importance values for the XGBoost classifier

Feature	Importance
PreInt_EduHx-computerinternet_hoursday	1.11
Physical-Height	1.09
FGC-FGC_CU	1.08
SDS-SDS_Total_Raw	1.07
Basic_Demos-Sex	1.05
Physical-Waist_Circumference	0.99
Physical-Weight	0.98
Fitness_Endurance-Max_Stage	0.97
FGC-FGC_TL	0.97
FGC-FGC_PU	0.95
CGAS-CGAS_Score	0.95
BIA-BIA_BMC	0.94
FGC-FGC_SR_Asymmetry	0.92
FGC-FGC_GSND	0.92
Physical-Systolic_BP	0.92
Fitness_Endurance-Time_Total_Sec	0.91
BIA-BIA_ICW	0.91
BIA-BIA_Activity_Level_num	0.91
Physical-HeartRate	0.90
FGC-FGC_SR_Mean	0.89

Table 13 reports the global XGBoost feature importance, defined as the average gain across all splits in which a feature is used. The comparison with the SHAP beeswarm plot in Figure 18 shows both convergences and discrepancies between the global gain-based importance and the class-specific global SHAP importance for **Moderate+Severe**. Some variables are consistently relevant in both perspectives: **PreInt_EduHx-computerinternet_hoursday** is the most important feature according to average gain and is also the second most influential SHAP feature for the Moderate+Severe class; similarly, **Physical-Height** and **Basic_Demos-Sex** appear among the most relevant variables in both rankings. This suggests that these features are structurally used by the boosted trees and also contribute substantially to the high-risk class output. However, **SDS-SDS_Total_Raw** shows a class-specific discrepancy: it is only fourth in the gain-based ranking, but it is the most important SHAP feature for Moderate+Severe, indicating that sleep-related impairment is especially informative for explaining the high-risk class rather than for the overall tree-building process alone. Conversely, features such as **FGC-FGC_CU** and **Physical-Waist_Circumference** rank highly in the global gain table but do not appear among the dominant SHAP contributors for Moderate+Severe, suggesting that they help the model split the data globally but are less decisive for this specific class. The opposite pattern is observed for **CGAS-CGAS_Score**, **BIA-BIA_ECW**, **Physical-HeartRate**, and **Fitness_Endurance-Time_Total_Sec**, which have stronger SHAP relevance for the high-risk output than their gain ranking would suggest. Overall, this comparison confirms the importance of combining model-internal global importance with class-specific

SHAP explanations.

The analysis then focuses on explaining the classifier’s decision for **three specific records**, one for each class. Their predicted probabilities were **not equally confident**: the None instance was classified with high confidence, the Mild instance was close to the decision boundary, and the Moderate+Severe instance was even more uncertain. This suggests that the high-risk class is not only underrepresented, but also weakly separated from the other classes in the learned feature space.

Explained instance	Predicted class	Predicted probabilities
None example	None (0)	None=0.92, Mild=0.08, Moderate+Severe=0.00
Mild example	Mild (1)	None=0.43, Mild=0.55, Moderate+Severe=0.02
Moderate+Severe example	Moderate+Severe (2)	None=0.36, Mild=0.24, Moderate+Severe=0.41

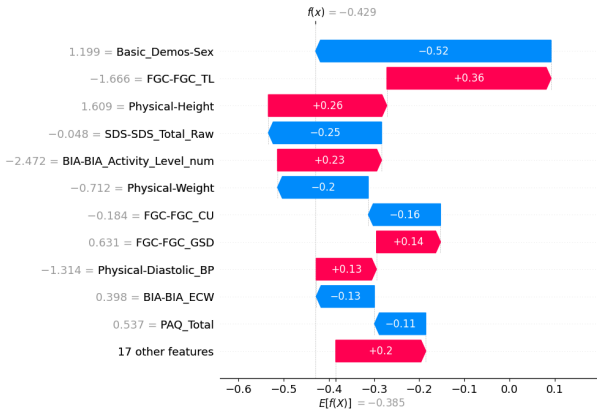


Figure 19: SHAP waterfall plot for the Moderate+Severe instance

Figure 19 present the waterfall plot of the features that most influenced the model’s decision for the correctly classified **Moderate+Severe case**, which is the selected instance with the lowest predicted-class probability (0.41). The largest absolute SHAP contributions for the predicted class are **Basic_Demos-Sex** (-0.521), **FGC-FGC_TL** (+0.363), **Physical-Height** (+0.262), **SDS-SDS_Total_Raw** (-0.251), and **BIA-BIA_Activity_Level_num** (+0.230). This local pattern is informative because it does not simply replicate the global ranking: although SDS is globally the most important high-risk feature, in this particular instance its standardized value is close to the mean and **contributes negatively** to the high-risk score.

The instance is therefore classified as Moderate+Severe by a combination of other **local signals**, and the predicted probability remains low.

6.2 LORE

While SHAP decomposes the model output into additive feature contributions, **LORE** builds a local **surrogate decision tree** around a synthetic neighborhood of the instance and extracts a rule explaining the black-box prediction. It also provides **counterfactual rules** showing how the local conditions would have to change to obtain a different predicted class. Since the input data were standardized, the thresholds in LORE rules are z-score thresholds.

Instance	Predicted class	Fidelity	Rule highlights	First counterfactual
None	None (0)	0.9946	SDS $\leq$ 0.77; FGC_CU interval; SR mean; BIA fat; BP; ECW.	No counterfactual returned in the notebook output.
Mild	Mild (1)	0.9944	SDS $\leq$ 0.63; height interval; endurance time; systolic BP; CGAS $\leq$ -0.94.	CGAS $>$ -0.94 $\rightarrow$ prediction changes to None.
Moderate+Severe	Moderate+Severe (2)	0.9891	PreInt hours $>$ 0.58; SDS $\leq$ 0.02; PreInt hours $\leq$ 0.58 $\rightarrow$ pre-ECW range; low diastolic BP; height/weight constraints.	PreInt hours $\leq$ 0.58 $\rightarrow$ prediction changes to None.

Table 14: Summary of the LORE local explanations for the three selected instances.

The most relevant LORE result is the counterfactual generated for the Moderate+Severe instance that initially is predicted as Moderate+Severe, but after applying the first counterfactual the black-box prediction becomes **None**. Only one feature is effectively changed: **PreInt_EduHx-computerinternet_hoursday**, LORE identifies a local **surrogate threshold** corresponding to 1.58 in the original scale, but this value has the discrete domain

0,1,2,3. The nearest admissible value satisfying the counterfactual rule is therefore 1.00, and direct evaluation of the XGBoost black box confirmed that changing the feature from 2.00 to 1.00 changes the prediction from Moderate+Severe to None.

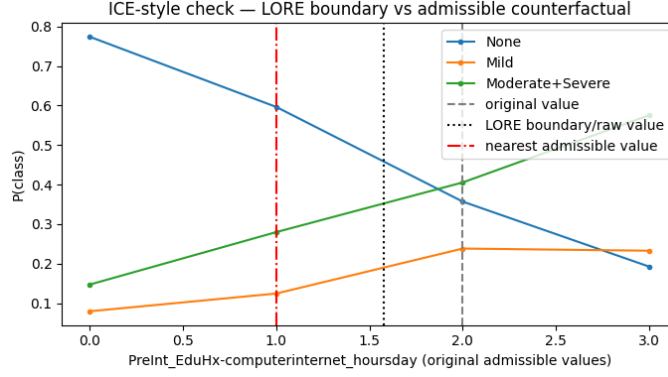


Figure 20: ICE-style check of the domain-valid LORE counterfactual for the Moderate+Severe instance

However, the LORE counterfactual should not be read as a causal statement such as “reducing internet hours would clinically remove risk” as it only states that, in the local approximation of this specific XGBoost decision boundary, changing this feature is sufficient to flip the model prediction. This distinction is supported by the SHAP check: for the same instance, `PreInt_EduHx-computerinternet_hoursday` contributes only **0.0806** to the high-risk output, accounting for 2.5% of the total absolute SHAP contribution and ranking 13th among the 28 features, therefore, SHAP and LORE do not completely agree locally. SHAP says that other features dominate the local score, whereas LORE identifies internet/computer hours as a sensitive threshold that can cross the decision boundary. This combination indicates that the instance lies in a locally sensitive region of the decision boundary and that the counterfactual must be presented as a local model-behavior explanation rather than as the unique reason for the classification.

However, this interpretation is conditional on the model and on the data used to train it. The poor recall of Moderate+Severe shows that these variables are **not sufficient** to separate the high-risk group reliably. In practice, the model appears conservative: it frequently predicts None unless the evidence for the minority classes is strong enough.

7 Data Understanding and Preparation of Time Series Dataset

7.1 Data Semantics and Time-Series Structure

The time-series dataset consists of **4,437** multivariate time-series instances, each containing **200** ordered observations, described as average signal values over **30-minute** intervals, and **13 columns**: acceleration along the three axes (*X*, *Y*, and *Z*), movement intensity (`enmo`), device orientation (`anglez`), ambient-light intensity (`light`), device-related information (`non-wear_flag` and `battery_voltage`), temporal information (`weekday`, `quarter`, and `relative_date_PCIAT`), the participant identifier (`id`), and the target (`sii_binary`), derived from the original `sii` by binarizing value 0 as class 0 and values 1, 2, and 3 as class 1. Although the dataset contains 4,437 time-series instances, they belong to only **494 unique participants**, with a mean of 8.98 instances per participant and a median of 2, ranging from 1 to 36. The target was consistent across all time series associated with the same participant.

7.2 Preliminary Data Quality Assessment

The exploratory analysis was divided between categorical (and contextual attributes) and continuous sensor signals. For the former, frequency distributions showed that **non-wear_flag**, despite being conceptually binary, contained 2,116 **intermediate values** between 0 and 1, corresponding to 0.24% of all observations. Moreover, **weekday**, **quarter**, and **relative_date_PCIAT** assumed a **single value** within each time series, which was not fully consistent with 200 consecutive 30-minute intervals, since these attributes would be expected to change over such a duration. Nevertheless, we accepted the 30 minutes temporal interpretation provided in the official dataset description.

For the continuous signals, graphical analysis of representative time series, distributions, and boxplots highlighted differences in scale and skewness, as well as the presence of potentially anomalous values. The correlation analysis also revealed a strong **redundancy** between **Z** and **anglez**, with a global Pearson correlation of approximately 0.993 and a mean within-series correlation of 0.988.

No missing values were detected in the dataset.

7.3 Data Cleaning: Our Choice

Following a conservative criterion, every non-zero **non-wear_flag** value was treated as non-wear. Although only 179 time series were affected, corresponding to 4.03% of the dataset, they involved 172 participants, or 34.82% of the total. However, 74.3% of the affected series contained non-wear observations for **more than half** of their length. Removing only these heavily contaminated instances would have retained 46 still partially contaminated time series but only 43 additional participants, while 125 participants would still have been completely excluded. We therefore **removed** every time series containing at least one non-wear observation, preserving uniform sequence length and avoiding potentially unreliable intervals. This reduced the dataset from 4,437 to **4,258** time series and from 494 to **326 participants**, since 168 participants had no completely clean time series available.

Potential outliers were examined using the IQR rule, the Hampel filter, and the Grubbs test. However, since wrist-sensor signals can naturally contain abrupt changes and movement peaks, unusual observations were considered plausible and retained. Only a few clearly **isolated extremes** visible in the distributions and boxplots were corrected. A conservative $3 \times \text{IQR}$ rule identified one value in **X**, one in **Y**, and three in **Z**. The same approach was not suitable for **enmo**, whose distribution was strongly right-skewed, so two isolated values of approximately 7.03 were manually identified as extreme, since the next-highest observed value was only about 2.52. The resulting seven removed cells were reconstructed through **linear interpolation** within their respective time series.

7.4 Attribute Reduction

The subsequent analyses retained only the five dynamic signals: {"X", "Y", "Z", "enmo", "light"}

The remaining attributes were **excluded** for the following reasons:

- **anglez** was removed because of its strong **redundancy** with **Z**;
- **battery_voltage** mainly described the **technical state** of the device rather than participant behaviour;
- **non-wear_flag** became constant after contaminated time series were removed and was therefore **uninformative**;
- **weekday**, **quarter**, and **relative_date_PCIAT** were excluded, as discussed above, because they were **constant** within each time series and did not represent evolving signals;
- **id** was used only to group observations by **participant**;
- **sii_binary** was retained exclusively as the **target** or for the interpretation of unsupervised results.

7.5 Standardization and Preservation of Temporal Dynamics

The retained signals had markedly different **numerical scales**, especially **light** compared with the accelerometer and movement channels. Where required by the downstream analysis, they were standardized channel-wise using z-score normalization to prevent variables with larger magnitudes from dominating distance-based analyses. The task-specific fitting procedures are described later in the clustering and classification sections.

No general detrending or smoothing was applied, since in this dataset gradual variations, abrupt movements, and local peaks may represent genuine behavioural or contextual information. Removing them could suppress patterns relevant to **clustering**, **motif and discord** discovery, and **classification**.

8 Motifs and Discords

Parameter Selection and Application: Motif and discord discovery was performed on **enmo**, the measure of movement intensity that provides an interpretable representation of **activity patterns**. The same signal was later used for the univariate shapelet analysis, allowing a direct comparison between unsupervised recurring or unusual patterns and supervised discriminative subsequences.

The preprocessed **enmo** signal was used without global standardization because the Matrix Profile comparison applies z-normalization separately to each subsequence. The resulting distances therefore primarily reflect similarities in local shape rather than differences in absolute signal magnitude.

A Matrix Profile was computed independently for each time series using STOMP. Two subsequence lengths were considered: $w = 6$ and $w = 12$, corresponding respectively to approximately 3 and 6 hours, to identify patterns throughout the day. Low Matrix Profile values were used to identify recurring patterns, or motifs, while high values were used to detect unusual subsequences, or discords.

For each time series and window length, up to three **motif** groups were extracted, each requiring at least two occurrences. An exclusion zone of $w/2$ was used to avoid treating strongly overlapping, slightly shifted versions of the same subsequence as distinct motif occurrences. Similarly, up to three **discords** were selected using an exclusion zone of w , ensuring that multiple selected discords represented distinct non-overlapping anomalous regions rather than shifted versions of the same event. Constant subsequences, for which local z-normalization is undefined, and other invalid Matrix Profile entries were excluded from the analysis.

A subsequent comparison with the discriminative shapelets identified during classification is presented after the classification analysis.

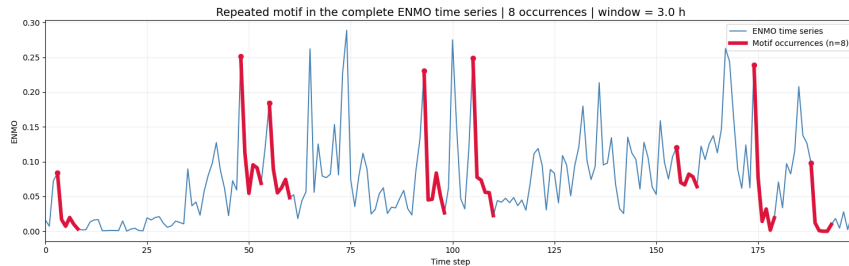


Figure 21: Example of a repeated motif in the complete ENMO time series. The highlighted red segments indicate eight occurrences of the same motif pattern with a window length of 3 hours.

Results: Motifs were detected in almost **all analyzable time series**. For $w = 6$, 2,850 series contained three motif groups and 1,329 contained two, while for $w = 12$ these numbers increased to 3,309 and 920, respectively. The strongest motif groups, defined by the **smallest distances** between their occurrences, were **not** necessarily the most **frequent**: the three strongest groups each contained only 2 occurrences, whereas the three most frequent

groups contained 14, 12, and 12 occurrences, respectively.

Up to three discords were explicitly selected for each time series and window length, and this limit was reached for almost every analyzable series.

The top 10 discords are characterized by repeated point-to-point **oscillations** between extremely negative and extremely positive values, producing an alternating **up-and-down pattern** across consecutive time points. This behaviour is representative of the most anomalous subsequences detected by the Matrix Profile analysis, where the discord score identifies temporal windows whose shape differs substantially from the rest of the same time series.

A preliminary comparison of aggregate Matrix Profile statistics across `sii_`-binary classes showed nearly identical values, providing **no evidence** of a direct association between overall local repetitiveness or uniqueness and the target.

Representative examples were then **inspected** to visualize both recurring patterns and locally unusual subsequences. The relationship between these unsupervised patterns and the discriminative shapelets identified during classification is discussed after the classification analysis.

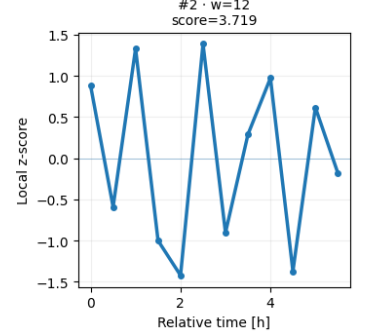


Figure 22: Example of a high-scoring discord subsequence extracted with window length $w = 12$.

9 Clustering

We performed clustering on the standardized dynamic signals using two feature configurations: one including `X`, `Y`, `Z`, `enmo`, and `light`, and one excluding `light` to also investigate clustering based only on **movement-related signals**. **K-Means** and Agglomerative **Hierarchical** Clustering were applied, while **PCA** and **t-SNE** were used for two-dimensional visualization.

Before clustering, each time series was reduced from 200 to 50 time steps through Piecewise Aggregate Approximation (**PAA**). Each PAA point therefore summarized four original observations, corresponding to approximately two hours. This **reduced dimensionality** and computational cost while preserving the temporal ordering and the main shape of the signals.

9.1 K-Means

Parameter Selection and Application: We used Euclidean distance. The number of clusters was selected by jointly considering the **silhouette score** and the reduction in **SSE**, seeking a compromise between cluster separation and the elbow of the SSE curve. With `light`, the main candidates were $k = 4$ and $k = 5$, with silhouette scores of 0.128 and 0.126, respectively. Since the SSE improvement became substantially smaller after $k = 4$, and after inspecting both solutions, **$k=4$** was selected. Without `light`, the main candidates were $k = 2$ and $k = 3$, with silhouette scores of 0.170 and 0.161. Since $k = 3$ clearly corresponded to the SSE elbow and provided a more informative partition while remaining close to the best silhouette value, **$k=3$** was selected.

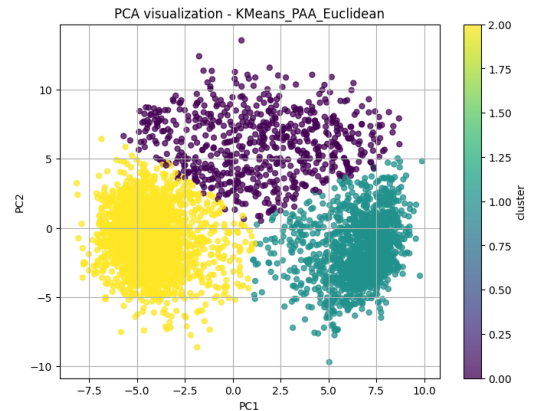


Figure 23: PCA visualization of K-Means clustering without the `light` signal.

Results: With **light**, K-Means produced **four** clusters ([240, 2150, 1241, 627]); their mean temporal profiles identified one small cluster characterized by markedly **higher enmo and light**, two large clusters mainly distinguished by **opposite X** patterns, and one cluster characterized by **high Z** values.

Without **light**, the three clusters ([652, 1349, 2257]) were mainly distinguished by a **high-Z** pattern and two **opposite X trajectories**. The removal of **light** therefore simplified the partition into three predominantly movement- and orientation-related profiles.

PCA and t-SNE confirmed the presence of these broad structures, although with some **overlap** between groups, consistently with the moderate silhouette values.

9.2 Hierarchical Clustering

Parameter Selection and Application: **Agglomerative Hierarchical Clustering** was applied using **Ward linkage** and **Euclidean distance**. The number of clusters was selected by jointly considering the **silhouette score** and inspection of the **dendrogram cut**.

With **light**, $k = 4$ and $k = 5$ were again the main candidates, with silhouette scores of 0.114 and 0.119. Despite the slightly higher score for $k = 5$, **k=4** was selected because the difference was minimal and the four-cluster solution was better supported by the dendrogram structure. Without **light**, $k = 2$, $k = 3$, and $k = 4$ obtained almost identical silhouette scores of 0.156, 0.155, and 0.156, respectively. Inspection of the hierarchical structure supported a three-cluster partition, and **k=3** was therefore selected.

Results: With **light**, four clusters were again obtained, ([253, 2212, 558, 1235]), with very similar distributions. After accounting for the arbitrary cluster labels, their temporal profiles closely matched the four **K-Means groups**: a high-enmo/high-light cluster, two groups with opposite X patterns, and a high- Z cluster.

Without **light**, the three clusters ([1451, 2321, 486]) again matched the K-Means structure, and the resulting profiles closely reproduced the same three patterns, but with **more overlap** between groups as seen in PCA and t-SNE.

9.3 Conclusion

The two algorithms produced highly **consistent qualitative partitions**. After accounting for arbitrary label permutations, both identified the same four main profiles when **light** was included and the same three

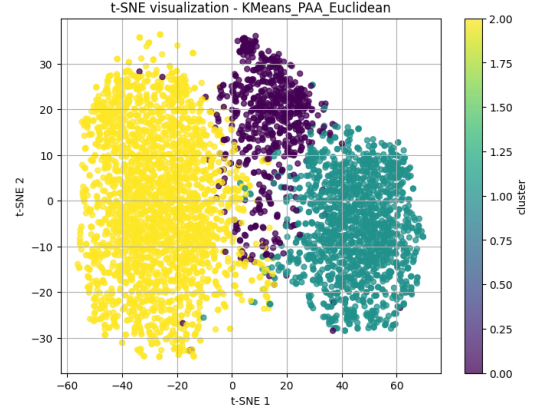


Figure 24: t-SNE visualization of K-Means clustering without the light signal.

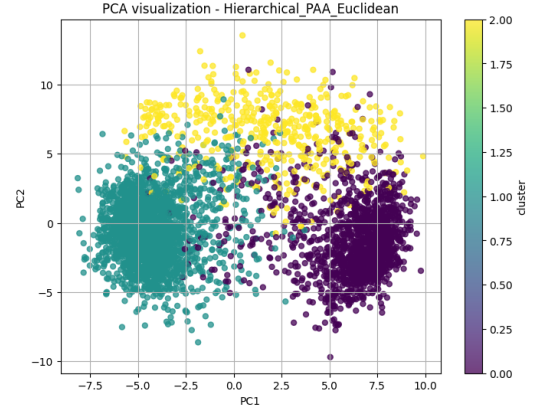


Figure 25: PCA visualization of hierarchical clustering without the light signal.

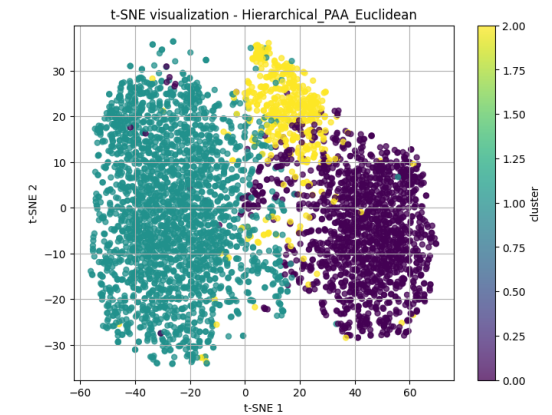


Figure 26: t-SNE visualization of hierarchical clustering without the light signal.

movement-related profiles when it was excluded. The similarity of both the temporal patterns and the corresponding cluster sizes suggests that the identified structure was not specific to a single clustering algorithm. K-Means provided slightly **better quantitative separation** in both selected configurations, with silhouette scores of 0.128 versus 0.114 with `light`, and 0.161 versus 0.155 without `light`. Hierarchical clustering nevertheless independently confirmed the same underlying organization. Removing `light` produced a **cleaner separation** according to the silhouette score and reduced the clustering to three main movement- and orientation-related patterns. Including `light`, however, revealed an **additional group** characterized by both higher ambient light and movement intensity, adding contextual information to the partition. Thus, the two configurations provide **complementary views** rather than one being universally preferable. Overall, the relatively moderate silhouette values and the overlap visible in PCA and t-SNE indicate that the dataset presents meaningful but not sharply separated cluster structure. A posteriori analysis of `sii_binary` across the resulting clusters did not reveal any clear association with the target, as both classes remained represented in all groups.

10 Classification

The **classification** task aimed to predict the binary target `sii_binary` using the five retained signals (`X`, `Y`, `Z`, `enmo`, and `light`). To avoid subject leakage, the dataset was split at participant level into training and test sets, so that no participant appeared in both partitions, into 70% training and 30% test sets. The signals were standardized channel-wise using parameters estimated exclusively from the training data and then applied to the test set; the same train-only procedure was repeated within each cross-validation fold. Since the target was moderately imbalanced at the time-series level (class 0: 67.05%, class 1: 32.95%), model selection focused primarily on **macro- F_1** rather than accuracy alone.

10.1 KNN with Euclidean and Manhattan Distance

Parameter Selection and Application: Each time series was flattened from shape (200, 5) into a 1,000-dimensional vector, preserving the point-to-point temporal correspondence between series. **Euclidean and Manhattan distances** were compared through a 5-fold **StratifiedGroupKFold**, preserving participant separation across folds. Odd values of k from 1 to 55 were tested (55 corresponds approximately to $\sqrt{3106}$, the square root of the number of training instances) together with both uniform and distance-based neighbour weighting. The best configuration was selected according to the cross-validated macro- F_1 score.

For Euclidean distance, the best configuration was **$k=1$** , making uniform and distance-based weighting equivalent, with a cross-validation macro- F_1 of 0.5117. For Manhattan distance, the

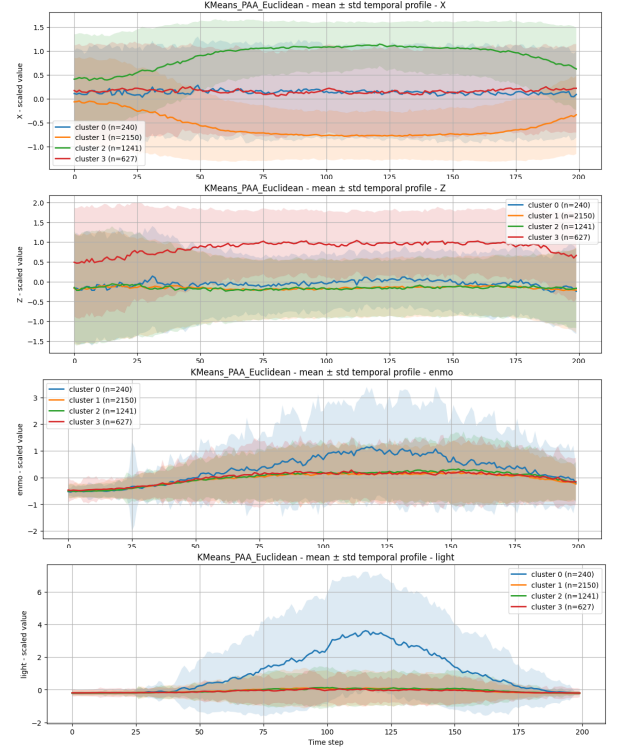


Figure 27: Mean temporal profiles with standard deviation bands for the K-Means clusters, considering the X, Z, ENMO and light signals.

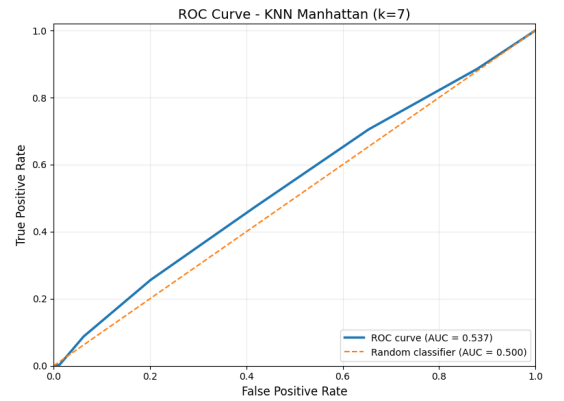


Figure 28: ROC curve for KNN with Manhattan distance.

best configuration was $k=7$ with uniform weights, with a slightly higher macro- F_1 of 0.512, making it the best overall configuration.

Results: The Euclidean model achieved an accuracy of 0.589 and a macro- F_1 of **0.514** on the test set. For the problematic class, it obtained a precision of 0.31, a recall of 0.34, and an F_1 -score of 0.32.

The Manhattan model performed slightly better overall, reaching an accuracy of 0.642 and a macro- F_1 of **0.526**. For the problematic class, it achieved a precision of 0.34, a recall of 0.26, and an F_1 -score of 0.29. The ROC curve remained only slightly above random performance, with an AUC of **0.537**.

Overall, **Manhattan distance** provided the best aggregate performance, while Euclidean distance achieved slightly better recall and F_1 -score for the problematic class. The two approaches therefore showed similar discriminatory ability, with neither providing a strong separation between the two target classes.

10.2 KNN with DTW

Parameter Selection and Application: Since **DTW** is considerably more computationally expensive than point-to-point distances, each time series was first reduced from 200 to 50 time steps through **PAA**. We tested odd values of k from 1 to 15 and both uniform and distance-based neighbour weighting using a 3-fold **StratifiedGroupKFold**. The best configuration was selected according to the cross-validated macro- F_1 score. The selected model used $k=1$, and achieved a cross-validation macro- F_1 of 0.502.

Results: On the test set, the DTW-based KNN achieved an accuracy of 0.564 and a macro- F_1 of **0.475**. For the problematic class, it obtained a precision of 0.25, a recall of 0.27, and an F_1 -score of 0.26. The ROC curve yielded an AUC of **0.475**, indicating no effective discrimination beyond random performance. Results obtained with larger values of k showed substantially higher accuracy, but lower macro- F_1 , indicating a stronger **bias** toward the majority class. Despite allowing nonlinear temporal alignment, DTW did not improve classification performance over the simpler Euclidean and Manhattan distances, suggesting that temporal misalignment alone was not the main factor limiting class discrimination.

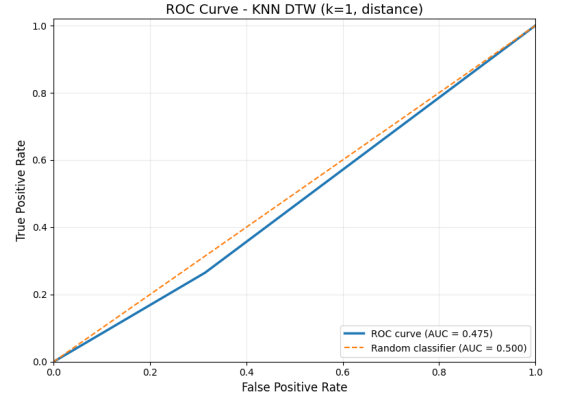


Figure 29: ROC curve for KNN with DTW distance.

10.3 Other Method: MiniROCKET

Parameter Selection and Application: **MiniROCKET** (RocketClassifier with rocket_transform="minirocket") transforms the original time series into a high-dimensional feature representation through a large set of convolutional kernels, which is then used for classification, and our selected configuration used **10,000 kernels**.

Results: This model achieved an accuracy of 0.638 and a macro- F_1 of **0.552**. For the problematic class, it obtained a precision of 0.37, a recall of 0.35, and an F_1 -score of 0.36. The ROC curve showed modest discrimination, with an AUC of **0.548**. Additional experiments with a larger MiniROCKET

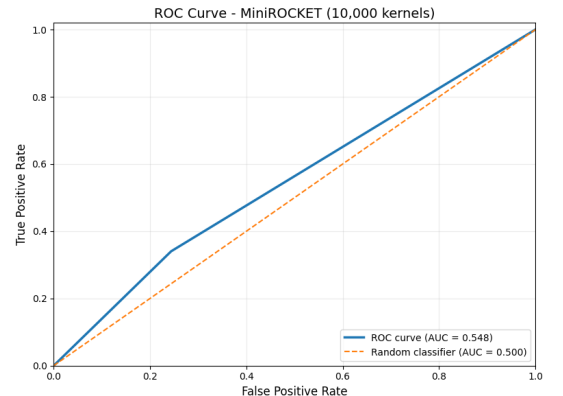


Figure 30: ROC curve for MiniROCKET with 10,000 kernels.

configuration and standard ROCKET did not improve these results; in particular, ROCKET achieved a lower accuracy of 0.602 and macro- F_1 of **0.517**. MiniROCKET therefore provided a relatively balanced compromise between overall accuracy and recognition of the minority class.

10.4 Shapelet-Based Classification

Parameter Selection and Application: We evaluated three shapelet-based configurations. Two **univariate approaches** were applied to **enmo**, both evaluating 1,000 candidate shapelets and retaining at most 10 in an initial computationally lighter setting and subsequently 100 to assess whether a richer shapelet representation improved performance: a **Shapelet Transform Classifier** and a **Random Shapelet Transform followed by Logistic Regression**, with candidate lengths ranging from 6 to 40 time steps in the latter. Finally, a **multivariate shapelet classifier** (still on the Shapelet Transform Classifier) was tested on all five signals (*X*, *Y*, *Z*, **enmo**, and **light**). To reduce computational cost, the series were first compressed from 200 to 50 time steps through PAA, after which 200 candidate shapelets were evaluated, again comparing retention limits of 10 and 100.

Results: Increasing the maximum number of retained shapelets from 10 to 100 improved accuracy for all three shapelet-based approaches, but **improved macro- F_1** only for the Random Shapelet Transform followed by Logistic Regression, while reducing it for the other two methods. We therefore report the best results for each method according to macro- F_1 , which was used as the primary evaluation criterion.

The univariate Shapelet Transform Classifier achieved the highest accuracy among the shapelet-based models, equal to 0.694, with a macro- F_1 of **0.523**, but it performed poorly on the problematic class, whose F_1 -score was only 0.24.

The Random Shapelet Transform followed by Logistic Regression achieved a lower accuracy of 0.608 but the **best shapelet-based macro- F_1** , equal to **0.567**, with substantially better recognition of the problematic class: precision 0.37, recall 0.52, and F_1 -score 0.43. The ROC curve achieved an AUC of **0.599**, the **highest** among the reported time-series classifiers. The Logistic Regression coefficients were also used to rank the extracted shapelets according to their contribution to the classification.

The multivariate shapelet classifier achieved an accuracy of 0.678 but a lower macro- F_1 of **0.495**, with an F_1 -score of only 0.19 for the problematic class. Overall, adding all five channels did not improve class discrimination, while the Random Shapelet Transform on **enmo** provided the best balance between the two classes and the most interpretable set of discriminative subsequences. The corresponding shapelets were therefore retained for the subsequent comparison with motifs and discords.

10.5 Conclusion

Overall, the **Random Shapelet Transform** followed by Logistic Regression achieved the best classification performance in terms of macro- F_1 (0.567), narrowly outperforming MiniROCKET (0.552) while providing sub-

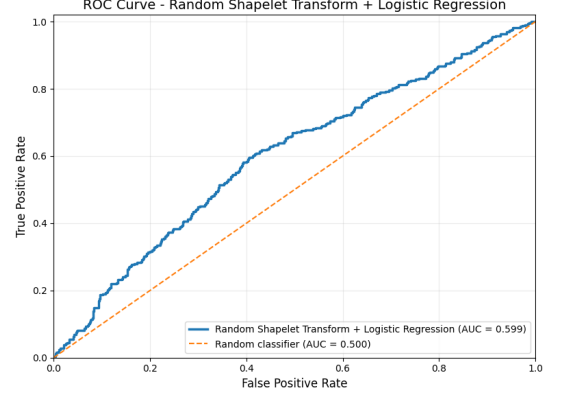


Figure 31: ROC curve for Random Shapelet Transform followed by Logistic Regression.

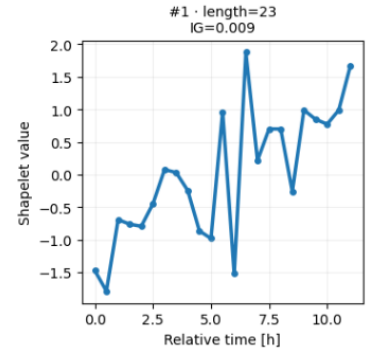


Figure 32: Example of the most informative shapelet extracted from the ENMO signal.

stantially **better recognition of the problematic class** (F_1 -score 0.43 vs. 0.36) and a more interpretable set of discriminative subsequences. In contrast, DTW-based KNN produced the weakest overall result, indicating that nonlinear temporal alignment alone was not sufficient to improve class separation. Although every evaluated method achieved accuracy values comparable to or lower than the trivial baseline of always assigning the most frequent class, the analysis still provides value because **it focused specifically on recognizing the minority class**; macro- F_1 was deliberately used as the primary evaluation criterion rather than accuracy for this specific reason.

11 Final Comparison Shapelets - Motifs/Discords

Comparison Method: The **top 10 shapelets** retained by the best Random Shapelet Transform model were compared with the strongest **motifs** (those with the smallest distances between their occurrences, rather than the most frequent groups) and **discords** identified by the Matrix Profile analysis. For each window length, $w = 6$ and $w = 12$, the 10 globally strongest motif groups and the 10 strongest discords were selected. Motifs were ranked by increasing motif distance, while discords were ranked by decreasing discord score.

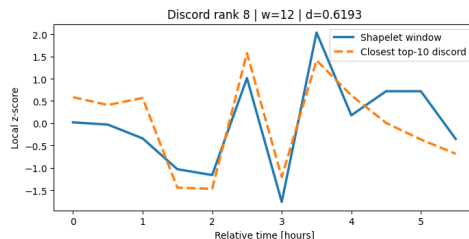


Figure 33: Comparison between a selected shapelet window and its closest top-10 discord subsequence.

Since the retained shapelets had lengths between 23 and 39 time steps, each shapelet was divided into all possible contiguous subsequences of length 6 and 12. Each resulting subsequence was then compared with patterns of the same length using z-normalized **Euclidean distance**. For motif groups, all occurrences belonging to the selected groups were considered. For each shapelet and window length, the smallest distance to a motif and to a discord was retained.

Results: At both temporal scales, 9 of the 10 shapelets were **closer to a top-ranked motif** than to a top-ranked discord, with the same single shapelet favouring a discord for both $w = 6$ and $w = 12$. For $w = 6$, the mean minimum distance was 0.649 for motifs and 1.261 for discords, with corresponding medians of 0.574 and 1.200. For $w = 12$, the mean distances were 1.559 and 3.025, with medians of 1.426 and 3.164, respectively. Since distance values depend on subsequence length, the two temporal scales were interpreted separately.

Extending the comparison from the top-ranked patterns to all extracted motif occurrences and discords produced substantially **smaller minimum distances**, as expected from the larger candidate pool. For $w = 6$, the mean distance decreased to 0.067 for motifs and 0.124 for discords, with 7 shapelets closer to a motif and 3 to a discord. For $w = 12$, the corresponding means were 0.196 and 0.826, with 9 shapelets closer to a motif and only one to a discord.

Overall, the discriminative shapelets showed a stronger similarity to recurring **local patterns** than to anomalous subsequences. This tendency was particularly clear among the globally strongest patterns at both temporal scales, and remained evident in the unrestricted comparison, especially for $w = 12$. The less marked distinction at $w = 6$ indicates that some shapelets also closely resembled discord-like patterns when the full set of extracted subsequences was considered.